

MoodDoctor: An Emotionally Intelligent Multimodal Support Companion built on AWS Bedrock

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Context: Built for the *AI for Bharat* Hackathon

Abstract

In the contemporary digital era, accessibility to mental wellness support remains a significant challenge. **MoodDoctor** is an advanced, emotionally intelligent support companion designed to bridge this gap. By leveraging multimodal sentiment analysis—integrating text, facial expressions, and vocal tones—MoodDoctor provides nuanced, context-aware emotional guidance. Powering this experience is **Amazon Bedrock**, which enables high-performance, secure, and scalable AI inference. This paper details the architecture, methodology, and implementation of MoodDoctor as a transformative tool for personalized mental wellness.

1 Introduction

The “AI for Bharat” initiative emphasizes the creation of AI solutions that are inclusive, accessible, and culturally resonant. Mental health, often underserved by traditional platforms, requires a more empathetic approach than standard LLM chatbots can provide. MoodDoctor was conceived to move beyond simple text interaction, creating a “Safe Space” that understands user emotions through multiple sensory inputs.

2 System Architecture

The system employs a modern, decoupled client-server architecture optimized for low-latency AI interactions and secure data management.

2.1 Frontend: The Interactive Interface

- **Core Stack:** Developed using **React 19 (Vite)** and **TypeScript**, ensuring a robust and type-safe user experience.
- **Sensory Capture:** Utilizes custom hooks for real-time webcam frame processing and audio recording.
- **Visual Analytics:** Implements **Recharts** to visualize mood trends, providing users with tangible insights into their emotional journey.

2.2 Backend: The Orchestration Layer

- **Framework: FastAPI (Python 3.12)** serves as the high-performance backbone, managing asynchronous API requests.
- **Database: MongoDB** stores session archives and personalized safety plans, ensuring continuity of care.
- **Security:** JWT-based authentication ensures user privacy and data integrity.

2.3 AI Integration: Amazon Bedrock

The project underwent a significant pivot to **Amazon Bedrock** to meet enterprise-grade scalability and hackathon requirements.

- **Models:** Leveraging **Claude 3 (Sonnet/Haiku)** and **Llama 3** for multimodal reasoning.
- **Interface:** Integrated via the **AWS SDK (Boto3)**, allowing for seamless prompting and image-to-text analysis.

3 Methodology: Multimodal Sentiment Analysis

MoodDoctor’s core innovation lies in its ability to synthesize three distinct streams of emotional data:

1. **Textual Analysis:** Deep semantic understanding of the user’s language to detect subtle emotional cues.
2. **Facial Expression Analysis:** Multimodal models analyze image frames captured via the webcam to identify visual markers of distress or joy.
3. **Vocal Tone Analysis:** (Ongoing) Processing audio energy and pitch to complement the visual and textual data.

3.1 Emotionally Adaptive Support Modes

Based on the synthesized data, the system dynamically switches between specialized “Support Modes”:

- **LISTENING:** Reflective mirroring for neutral states.
- **CALMING:** Grounding techniques for high-anxiety detection.
- **CRISIS__AWARE:** Immediate activation of safety protocols if high risk is detected.

4 Key Features

- **Therapy Sanctuary:** An immersive chat environment with “Biometric Sync” that provides visual feedback of the user’s detected mood.
- **Safety Planning:** A dedicated module for users to document coping strategies, accessible instantly during periods of distress.
- **Session Archiving:** Encrypted storage of past interactions to foster long-term reflection.

5 Implementation Highlights: The AWS Pivot

The transition to **Amazon Bedrock** allowed for a unified AI pipeline. By using a single API for both text generation and image-based sentiment analysis, we reduced latency and simplified the backend logic. The use of **Boto3** ensured that our infrastructure follows industry-standard cloud practices.

6 Conclusion and Future Scope

MoodDoctor demonstrates that AI can be more than just a tool; it can be an empathetic companion. By combining multimodal inputs with the power of AWS Bedrock, we have created a scalable blueprint for AI-driven mental wellness. Future iterations will focus on localizing the experience for different Indian languages, aligning with the “AI for Bharat” vision.

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